

PINAR DEL RIO, Cuba

WMO: 783150

Lat: 22.417N

Lon: 83.683W

Elev: 37

StdP: 100.88

Time Zone: -5.00 (CUB)

Period: 82-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	14.5	9.0	7.2	16.9	10.3	7.8	17.9	8.4	23.1	7.7	23.2	0.4	40	0.243

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	32.5	25.1	32.0	25.0	31.4	24.9	26.9	30.3	26.5	30.0	26.1	29.7	2.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	29.0	25.5	20.8	28.5	25.1	20.3	28.2	84.5	30.4	82.6	30.0	81.1	29.8	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.2	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	21.7	22.5	23.2	25.0	26.2	27.0	27.3	27.5	26.9	25.6	24.0	22.1
	DBStd	2.63	2.09	2.16	2.22	1.59	1.10	1.07	1.02	0.93	0.91	1.38	1.89	2.24
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	8	3	2	1	0	0	0	0	0	0	0	0	3
	CDD10.0	5443	363	349	408	450	503	509	536	542	506	482	421	374
	CDD18.3	2409	107	117	150	200	245	259	278	284	256	224	171	119
	CDH23.3	18324	502	671	1033	1569	1998	2281	2491	2563	2046	1593	1012	564
	CDH26.7	5488	53	118	234	473	684	748	872	938	651	439	212	66
Wind	WSAvg	1.5	1.6	1.7	1.9	1.7	1.6	1.4	1.2	1.1	1.1	1.4	1.6	1.6
Precipitation	PrecAvg	1397	64	45	55	80	148	214	140	163	211	161	57	46
	PrecMax	1946	228	85	134	220	355	580	281	302	457	333	179	161
	PrecMin	1024	4	12	2	18	41	75	81	102	86	46	2	3
	PrecStd	249	63	21	35	49	80	136	49	63	85	83	41	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.3	30.0	31.0	31.8	32.7	32.7	33.1	33.7	32.5	31.7	30.6	29.1
		MCWB	22.3	22.2	23.4	22.9	23.4	25.3	25.4	25.2	25.2	25.0	24.0	22.4
	2%	DB	28.1	29.0	29.8	30.9	31.7	31.8	32.2	32.5	31.7	30.7	29.7	28.2
		MCWB	22.0	21.9	22.9	22.9	23.3	25.3	25.3	25.6	25.3	24.7	23.7	22.2
	5%	DB	27.3	28.1	29.0	30.3	30.9	31.1	31.5	31.8	31.0	30.1	29.0	27.4
		MCWB	21.6	21.8	22.5	22.8	23.4	25.4	25.3	25.5	25.4	24.7	23.6	21.9
	10%	DB	26.4	27.1	28.1	29.4	30.1	30.4	30.8	31.0	30.2	29.3	28.1	26.5
		MCWB	21.2	21.6	22.1	22.7	23.4	25.3	25.3	25.6	25.3	24.2	23.4	21.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.1	25.1	25.2	26.3	27.6	27.2	27.6	26.7	26.4	25.5	24.1
		MCDB	27.1	27.6	29.1	29.0	29.9	30.1	30.7	31.3	30.0	29.6	28.4	27.1
	2%	WB	23.2	23.5	24.2	24.6	25.6	26.7	26.6	26.9	26.3	25.7	24.8	23.2
		MCDB	26.2	26.9	28.0	28.6	29.3	29.7	30.2	30.6	29.8	28.9	28.0	26.3
	5%	WB	22.6	22.9	23.5	24.1	25.1	26.2	26.1	26.3	26.0	25.4	24.3	22.7
		MCDB	25.6	26.2	27.3	28.1	28.6	29.5	29.9	30.2	29.6	28.5	27.6	25.8
	10%	WB	22.0	22.3	23.0	23.6	24.5	25.8	25.7	25.9	25.7	24.9	23.9	22.2
		MCDB	25.2	25.6	26.7	27.5	27.9	29.1	29.4	29.7	29.1	28.0	27.1	25.2
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.4	9.4	9.1	8.1	7.0	7.5	7.8	7.2	7.4	7.9	8.6
		MCDBR	9.6	9.8	9.7	9.5	9.2	8.1	8.4	8.6	8.3	8.3	8.2	8.8
		MCWBR	4.4	4.3	4.0	3.3	3.0	3.1	3.1	3.3	3.3	3.3	3.3	3.8
	5% WB	MCDBR	8.3	8.8	9.1	8.6	7.6	6.8	7.6	7.8	7.3	7.0	7.4	7.8
		MCWBR	4.6	4.3	4.6	4.0	3.3	3.3	3.5	3.7	3.3	3.3	3.6	4.0
Clear-Sky Solar Irradiance	taub		0.365	0.370	0.375	0.406	0.431	0.450	0.466	0.433	0.417	0.401	0.373	0.367
	taud		2.518	2.511	2.478	2.389	2.365	2.359	2.303	2.423	2.453	2.489	2.514	2.539
	Ebn at Noon		903	921	928	898	867	844	831	862	874	878	887	888
	Edh at Noon		97	103	111	123	125	125	132	118	112	104	96	92
All-Sky Solar Radiation	RadAvg		4.00	4.84	5.77	6.39	6.14	5.42	5.99	5.72	5.13	4.67	4.17	3.83
	RadStd		0.30	0.33	0.23	0.23	0.52	0.41	0.28	0.20	0.33	0.37	0.22	0.28

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page